

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHMS

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DAA LAB EXPERIMENT 3

Bubble Sort with Big-O Analysis

Aim

To implement Bubble Sort in Python and analyze its time and space complexity using Big-O notation.

Algorithm

1. Start.
2. Read an array of elements.
3. Compare adjacent elements.
4. If the first element is greater than the second, swap them.
5. Continue the process for all adjacent elements.
6. Repeat the passes until the array is sorted.
7. Display the sorted array.
8. Stop.

Python Source Code

```
def bubble_sort(arr):  
    n = len(arr)  
    for i in range(n):  
        swapped = False  
        for j in range(0, n - i - 1):  
            if arr[j] > arr[j + 1]:  
                arr[j], arr[j + 1] = arr[j + 1], arr[j]  
                swapped = True  
        if not swapped:  
            break
```

```
arr = [64, 34, 25, 12, 22, 11, 90]
```

```
print("Original Array:", arr)
```

```
bubble_sort(arr)
```

```
print("Sorted Array:", arr)
```

Output

main.py	Visualize	Run ▶	Output
<pre>1 def bubble_sort(arr): 2 n = len(arr) 3 4 for i in range(n): 5 swapped = False 6 7 for j in range(0, n - i - 1): 8 if arr[j] > arr[j + 1]: 9 arr[j], arr[j + 1] = arr[j + 1], arr[j] 10 swapped = True 11 12 if not swapped: 13 break 14 15 16 arr = [64, 34, 25, 12, 22, 11, 90] 17 18 print("Original Array:", arr) 19 20 bubble_sort(arr) 21 22 print("Sorted Array:", arr)</pre>			<pre>Original Array: [64, 34, 25, 12, 22, 11, 90] Sorted Array: [11, 12, 22, 25, 34, 64, 90]</pre>

Result

Thus, the **Bubble Sort algorithm** was successfully implemented in Python, and its best, average, and worst-case complexities were analyzed.